Gaussian context sampling for imbalanced regression with tabular in-context learners

TODO: authors, affiliation

Table 1: TabICL, synthetic-prior continued pre-training only. No model has seen any evaluation dataset. Mean ± sd over 5 seeds; lower is better.

	Many	Medium
Abalone		
zero-shot	1.146 ± 0.081	2.595 ± 0.031
uniform context	1.148 ± 0.082	$\mathbf{2.578 \pm 0.031}$
target-balanced context	*TODO*	*TODO*
Gaussian context	$\mathbf{1.128 \pm 0.087}$	2.618 ± 0.024
Communities		
zero-shot	0.0587 ± 0.0013	$\mathbf{0.1680 \pm 0.0054}$
uniform context	0.0585 ± 0.0014	0.1683 ± 0.0055
target-balanced context	*TODO*	*TODO*
Gaussian context	$\mathbf{0.0577 \pm 0.0015}$	0.1737 ± 0.0066
Wine Quality White		
zero-shot	$\mathbf{0.335 \pm 0.008}$	$\mathbf{1.195 \pm 0.060}$
uniform context	0.337 ± 0.009	1.199 ± 0.059
target-balanced context	*TODO*	*TODO*
Gaussian context	0.349 ± 0.011	1.280 ± 0.051

Table 2: TabPFN-2.5, per-dataset fine-tuning. Learning rate selected on validation.

	Few	Many Overall	Medium GM	Fe
Abalone				
zero-shot	*TODO*	*TODO*	*TODO*	*TO*
uniform fine-tuning	6.219 ± 0.030	$\mathbf{1.208 \pm 0.086}$	$\mathbf{2.057 \pm 0.085}$	$5.767 \pm$
Gaussian fine-tuning	$\mathbf{6.189 \pm 0.031}$	1.943 ± 0.059	$\mathbf{1.692 \pm 0.074}$	$\mathbf{4.127} \pm$
Communities	*TODO*	*TODO*	*TODO*	
zero-shot	6.385 ± 0.037	3.000 ± 0.028	1.665 ± 0.075	
uniform fine-tuning	0.0597 ± 0.0016	$\mathbf{0.0592 \pm 0.0015}$	0.1677 ± 0.0049	$0.2577 \pm$
Gaussian fine-tuning	$\mathbf{0.0596 \pm 0.0017}$	0.0596 ± 0.0040	$\mathbf{0.1607 \pm 0.0042}$	$\mathbf{0.2348} \pm$
Wine Quality White	0.2549 ± 0.0082	0.1598 ± 0.0040	0.0896 ± 0.0072	
zero-shot	*TODO*	*TODO*	*TODO*	
uniform fine-tuning	0.2854 ± 0.0037	0.1708 ± 0.0031	0.0937 ± 0.0042	$2.455 \pm$
Gaussian fine-tuning	$\mathbf{0.370 \pm 0.019}$	0.391 ± 0.038	1.295 ± 0.023	$2.375 \pm$
	$\mathbf{2.453 \pm 0.024}$	$\mathbf{0.785 \pm 0.020}$	$\mathbf{1.138 \pm 0.055}$	$\mathbf{2.250} \pm$
	2.456 ± 0.025	0.787 ± 0.019	0.256 ± 0.008	

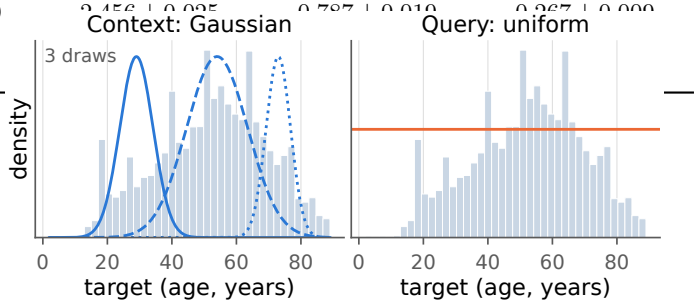


Figure 1: Context rows are weighted by a Gaussian over the target, with centre and width redrawn each episode; queries stay uniform.

Abstract

1 Introduction

2 Method

2.1 Context and query sampling

Each episode draws a centre and a width,

$$\mu = Q_y(u), \quad u \sim U(0.05, 0.95), \qquad \sigma = \mathrm{sd}(y)\, e^{v}, \quad v \sim U(\log 0.15, \log 1.5), \tag{1}$$

and weights each candidate context row by $w_i \propto \exp\!\big(-\tfrac{1}{2}((y_i - \mu)/\sigma)^2\big)$.

2.2 Two regimes

2.3 ECG encoder

3 Experiments

3.1 Setup

3.2 Results

3.3 The trade-off

4 Conclusion and limitations

Table 3: PTB-XL age prediction, mean absolute error in years, on the official test split. Band-macro is the unweighted mean of the three band errors.

	<30	30–60	≥60	Natural	Band-macro	GM
ResNet (vanilla)	9.456 ± 0.000	$\mathbf{6.294 \pm 0.000}$	8.940 ± 0.000	7.548 ± 0.000	8.230 ± 0.000	4.904 ± 0.000
ResNet + RankSim + ConR	*TODO*	*TODO*	*TODO*	*TODO*	*TODO*	*TODO*
E2 + stock TabPFN-2.5	$\mathbf{7.657 \pm 0.009}$	7.245 ± 0.006	$\mathbf{7.628 \pm 0.011}$	$\mathbf{7.421 \pm 0.003}$	$\mathbf{7.510 \pm 0.005}$	$\mathbf{4.724 \pm 0.023}$
E2 + Gaussian TabPFN-2.5	7.657 ± 0.009	7.245 ± 0.006	7.628 ± 0.011	7.421 ± 0.003	7.510 ± 0.005	4.724 ± 0.023
E3 + stock TabICL	*TODO*	*TODO*	*TODO*	*TODO*	*TODO*	*TODO*
E3 + Gaussian TabICL	*TODO*	*TODO*	*TODO*	*TODO*	*TODO*	*TODO*

Table 4: Encoder training against prediction head. The gap between the two column groups is the contamination in the balanced protocol, not a property of the method.

	Natural (official split)		Balanced (DIR split)	
Encoder	stock head	Gaussian head	stock head	Gaussian head
E1 (supervised L1)	*TODO*	*TODO*	7.330 ± 0.067	7.162 ± 0.070
E2 (through TabPFN)	*TODO*	*TODO*	7.258 ± 0.090	7.108 ± 0.080
E3 (through stock TabICL)	*TODO*	*TODO*	$\mathbf{7.152 \pm 0.090}$	$\mathbf{7.042 \pm 0.120}$
E4 (through Gaussian TabICL)	*TODO*	*TODO*	7.384 ± 0.080	7.392 ± 0.108

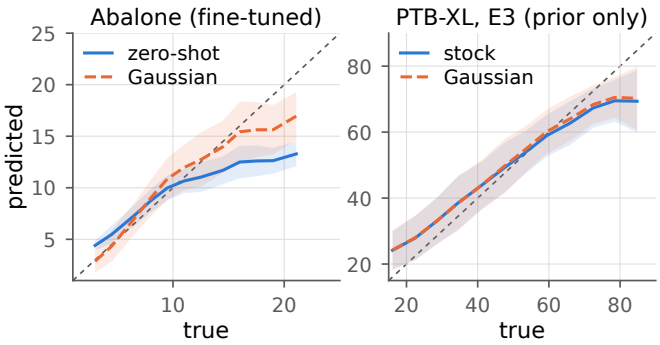


Figure 2: Binned predictions against truth, ±1 sd. TODO: state in the caption which arms and whether predictions are seed-averaged.